

1 A Fault-Resilient Measurement and Control Framework 2 for Controlled-Environment Hydroponics

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5 Abstract

The growing threat of climate change and global warming highlights the need for Controlled Climate Agriculture (CCA). Extreme weather patterns endanger global food supply, making hydroponics an emerging alternative to traditional farming. Automated hydroponic systems with Internet of Things (IoT) capabilities have been implemented worldwide, demonstrating their potential for small- and large-scale operations. Our proposed system builds on previous automated hydroponics work, monitoring, and controlling key physical parameters for optimal plant growth, with the novel addition of CO₂ gas supplementation and regulation. Two experimental iterations were used to test the efficacy of the system under different weather conditions. Using data from the first experiment, we trained and tested multiple machine-learning models to create an intelligent control action classifier. The XGBoostClassifier performed the best, achieving 0.98 accuracy, with 0.97 recall, 0.96 precision, and 0.96 F1-score. We also proposed and implemented a persistent error-resilience algorithm developed as a constrained optimisation problem. This involved a cost function and an adaptive multi-parameter λ -weighted minimization approach with update policies. Two experiments using green leaf lettuce compared our system with soil-based and manual Deep Water Culture (DWC) methods. The results showed a significantly higher fresh mass yield in our system. Our proposed system (P1) outperformed both soil-based (P2) and manual Deep Water Culture (DWC) methods (P3), with P1 producing 78.1% higher fresh mass yield than P2 and 168.8% higher than P3 in the first experiment. In the second experiment, P1's superiority was

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even more pronounced, with 288.1% higher yield than P2 and 141.9% higher than P3. Optimized on the first experimental data and implemented in the second, the error resilience algorithm reduced erroneous sensor data instances by approximately four times compared to the conventional threshold and bandpass filters used initially.

6 *Keywords:* Hydroponics, smart agriculture, IoT in agriculture,
7 water-quality sensors, intelligent control action classification, error resilience

8 **1. Introduction**

9 Hydroponics involves growing plants in aqueous nutrient solutions with or
10 without non-soil media. Commonly used media include expanded clay, coir,
11 perlite, vermiculite, brick shards, polystyrene packing peanuts, and wood
12 fiber [1]. This form of agriculture removes traditional dependence on arable
13 land, which is often a significant limiting factor for a region’s agricultural
14 output. Hydroponic systems are also ideal for smaller spaces, such as bal-
15 conies and rooftops, where it would otherwise be difficult to grow anything
16 via conventional methods [2]. Combining hydroponics with Controlled Envi-
17 ronment Agriculture (CEA) has the potential to revolutionize global agricul-
18 ture. Large-scale adaptations of this subset of agriculture would significantly
19 reduce the burden on existing arable land and cease searches for newer arable
20 lands, which leads to deforestation and ecological destruction [3], all the while
21 tackling a looming global food crisis [4].

22 Albeit past works have dived into developing an automated solution for
23 hydroponic systems, no single system manages to monitor and control all
24 the physical growth parameters necessary for the optimal growth of plants.
25 We have identified from past literature that water temperature, air temper-
26 ature, electrical conductivity, pH, Carbon Dioxide (CO₂) concentration, and
27 humidity contribute the most towards the growth of plants [5, 6, 7, 8]. Fur-
28 thermore, with IoT systems that run around the clock, there is always the
29 chance of erroneous readings from the sensors due to many reasons, ranging
30 from electrical fluctuations to sudden physical changes in the system. An au-
31 tomatic system might act on such erroneous data, leading to control actions
32 that can negatively affect plants’ growth and yield [9], [10].

33 Our contribution is motivated by the growing need for more efficient agri-
34 cultural techniques and imbuing them with automation to be easily moni-
35 tored and controlled remotely. The challenges of unreliable and erroneous

sensor readings also motivated us to address them, as they often plague IoT-enabled systems. In our work, we designed an automated hydroponics system, a closed, climate-controlled system where the aforementioned physical parameters known to contribute to plant growth can be regulated. To test the efficacy of our system, we grew a batch of 7 lettuce (*Lactuca sativa*) [11] plants against two equally sized batches of 7 lettuce plants: one batch grown in traditional soil treatment and the other in a Deep Water Culture (DWC) treatment. These two treatments served as our control groups. Intelligent classification methods were also explored intensively, which led us to implement an XGBoost-based control action determination method within the system to maintain a constant optimal environment for plant growth. The core contributions of this article are itemized as follows:

1. Based on past literature, we designed a Nutrient Film Technique (NFT) hydroponics system [12]. The system is closed to maintain a separate ambient environment inside of it, isolated from the outside. Equipping the system with numerous sensors and actuators, the water temperature, air temperature, electrical conductivity, pH, Carbon Dioxide CO₂ concentration, and humidity within the system is monitored and regulated. Among the physical growth parameters mentioned, CO₂ concentration was not monitored or controlled in any past literature that we had come across.
2. A Flask web application was developed to host a web-based dashboard. The sensor data from the system could be seen on the dashboard in real-time, along with visualizations to illustrate the trend in readings, allowing us to monitor and interpret the physical conditions within the system.
3. Using the sensor data collected from the batch of lettuce grown in the system, we explored numerous classification methods. Exploring and testing machine learning algorithms, we attempted to settle on the best approach for classifying the control actions required to maintain optimal physical conditions with the system for the best plant growth. We tried to develop an intelligent classification method that can effectively replace the conventional explicit conditional programming methods used in control action classification.
4. Automated systems that are designed to work around the clock are prone to acting on erroneous sensor data. For sensitive systems, such

as a hydroponics system where physical conditions create a "make-or-break" situation for plants, such erroneous actuation can damage plants or, in the worst-case scenario, kill them completely. To mitigate receiving and formulating decisions based on erroneous data from the sensors, we proposed an error resilience algorithm and tested it on our system. Applying a combination of filters allowed us to propose this method to effectively reduce the chances of erroneous values being passed onto the control action classifier.

The rest of the article is written as follows. Section 2 discusses past literature and related works, and Section 3 delves into the methodologies behind the design and implementation of the system. Section 4 and 5 cover the details of the two experiments performed with the proposed system against two control groups. Section 6 then carries on with the analysis to compare the results from the three treatments for lettuce growth along with the smart classification system. Finally, Section 7 summarizes our findings and offers suggestions for future work.

2. Related Work

Over the years, IoT has developed quite a foothold in agriculture. Going through past literature, we encountered numerous implementations of automated hydroponic systems with IoT technologies. In the following subsections, we discuss the various past related works involving the application of IoT and intelligent classification methods in hydroponics.

2.1. IoT in Hydroponics

The Internet of Things connects devices and people over the Internet, allowing data to be stored and analyzed online. This development has enabled farmers to control hydroponic systems automatically. With IoT, monitoring and adjusting water quality, temperature, and light for plant growth is now simpler and can be done remotely. There's been a notable increase in research focused on using IoT to manage and improve these soil-less farming systems. By implementing IoT, farmers can achieve more precise control and management of their hydroponic systems [13].

Sadek et al. [14] designed a smart greenhouse for hydroponics and aeroponics incorporating IoT, which was used for experimentation at the National

105 Water Research Centre in Egypt. The system effectively monitored air tem-
106 perature, humidity, luminous intensity, and TDS in the nutrient solution
107 supplied to the plants. They also developed a website where users can log in
108 to monitor the sensor readings from the system. The results of their study
109 showed that the plants matured in a period of 45 days, as opposed to 75
110 days, using traditional soil-based agricultural practices.

111 Sneineh and Shabaneh [15] has developed an IoT-enabled monitoring sys-
112 tem for a hydroponics system centred around an ESP32 microcontroller.
113 Their system can monitor Total Dissolved Solids (TDS), pH, water level,
114 and water temperature. The sensors send data to the ESP32, which then
115 performs actions based on the sensor data. Whenever the parameters ex-
116 ceeded the optimal range, the system automatically actuated pumps to add
117 nutrients and salts to nutrient solution to maintain optimal conditions for
118 plant growth. Blynk, a low-code IoT software platform, was used to develop
119 a mobile application where the sensor data could be seen, and the pumps
120 adding the nutrients and salt could be manually controlled if the farmers
121 decided to.

122 Mapari et al. [16] designed a Nutrient Film Technique (NFT) [17] hydro-
123 ponics system that monitors pH, humidity, TDS, air and water temperature
124 with a Node MCU microcontroller at its heart, which receives data from the
125 sensors and upload them to the mobile app server. A Blynk phone app was
126 developed to see the sensor readings from the system in real-time. A water
127 pump was used to control the in-flow of nutrient solution. The mobile app
128 notifies the user whenever the parameters are out of the desired range, and
129 the user decides whether to operate the water pump.

130 Tatas et al. [18] proposed iPONICS, an intelligent, low-cost IoT sys-
131 tem for hydroponic greenhouses. iPONICS utilizes a network of three sensor
132 nodes: a master node called the Sensing and Control Unit (SCU) centered
133 around an Arduino Mega 2560 that controls pumps, monitors water quality,
134 and aggregates data from slave nodes. Environment sensing slave nodes, col-
135 lectively called Environment Sensing Unit (ESU), track ambient conditions
136 within the greenhouse, while security nodes under the Greenhouse Security
137 Unit (GSU) monitor for activity. Their system can monitor water quality,
138 temperature, and humidity, ensuring optimal growing conditions. It also
139 controls irrigation based on a fuzzy inference engine and transmits data for
140 remote monitoring. Designed for low power consumption, iPONICS offers
141 potential for off-grid applications and fault tolerance.

142 Chowdhury et al. [19] designed an NFT hydroponic system that monitors

143 pH, humidity, electrical conductivity (EC), air temperature, and water flow
144 into the system from the nutrient solution tank. With an Arduino Mega as
145 the central processing unit, the sensor readings are used to determine the
146 control actions and pass them onto the actuators to implement the desired
147 changes within the system. They developed a dosing system to add acid or
148 base to adjust the pH and nutrient solution to adjust the concentration of
149 nutrients in the water supplied to the plants. An ESP 8266 Wi-Fi module is
150 used to upload the sensor data to ThingSpeak, an IoT analytics platform by
151 MathWorks that offers data visualization tools and widgets to visualize the
152 sensor data.

153 Domingues et al. [20] designed and developed an automated hydroponics
154 system specifically to grow lettuce. They proposed their system as an al-
155 ternative means for the mainstream agricultural production of lettuce. The
156 system monitors and controls pH, electrical conductivity, and water temper-
157 ature. They developed software named ControlHidro to monitor and control
158 the hydroponics system. Their proprietary software monitored the two pa-
159 rameters for 24 hours throughout the growth cycle of the lettuce. The pH
160 and EC were controlled by adding additional acid, base, and nutrients to the
161 nutrient solution by actuating 2-way solenoid valves. They also grew a batch
162 of lettuce in soil using traditional methods as a control group. They ana-
163 lyzed the agricultural and nutritional values of the lettuce grown using their
164 system against that of the control group, which showed that both methods of
165 agriculture yielded comparable results, highlighting that hydroponics is just
166 as viable as soil-based methods.

167 2.2. *Artificial Intelligence in Hydroponics*

168 Over the years, Artificial Intelligence (AI) has emerged as a turning point
169 in almost every field, from electricity-run technology to life-saving medicine
170 to even fields as profoundly intricate and human-esque as the arts. Within
171 the expansive landscape of AI is machine learning, a subset of this growing
172 sector that empowers computers to execute tasks autonomously after being
173 trained for specific activities using a considerable amount of data. To sim-
174 ulate human thinking, a machine must first learn and process information
175 like human cognition. Humans make decisions based on past experiences and
176 data, and similarly, machine learning algorithms use historical data to inform
177 future actions. In hydroponics, these algorithms are crucial in optimizing the
178 physical conditions inside hydroponic systems.

179 Mehra et al. [21] designed and developed an intelligent IoT-enabled hy-
180 droponics system that uses deep neural networks to determine the control
181 actions required to adjust the physical conditions within the growth space of
182 the system where the team grew tomatoes. This system monitors humidity,
183 water level, and light intensity. The system was designed to take readings
184 from the system using sensors that send the data over to the microcontroller,
185 which transfers the data to the Raspberry Pi 3. The deep neural network
186 model is loaded onto the Raspberry Pi 3, which uses the sensor readings to
187 predict the control action needed. They grew a single tomato plant in their
188 system and compared it against another plant planted simultaneously as a
189 control. They compared the height of the two plants throughout 30 days,
190 showing that the plants grown in their system grew significantly faster than
191 the soil-based approach.

192 Adidrana et al. [22] designed an automated IoT-enabled NFT hydro-
193 ponics system where they measured pH, EC, water temperature, and TDS.
194 They then applied the k-Nearest Neighbor (KNN) algorithm [23], a super-
195 vised learning classifier, to predict the classification of nutrient conditions.
196 The microcontroller takes the predicted results from the KNN algorithm. It
197 sends commands to the actuator to vary the conditions of the nutrient so-
198 lution and the system to maintain an optimal environment. They achieved
199 an accuracy of 93.3% in predicting the correct control action to be taken by
200 their system.

201 Nugroho et al. [24] developed an NFT hydroponics system capable of
202 measuring water temperature, water level, and humidity. They collected
203 data from this system and corresponding control actions from it. Using 4,000
204 such data points, they trained a Deep Neural Network (DNN) to predict
205 the required control action given real-time sensor readings as input. Their
206 model achieved an accuracy of 81% after training, and they later tested it on
207 their system to evaluate its real-world performance in classifying the correct
208 control action based on sensor data.

209 Therefore, our work attempts to monitor and control all the physical
210 growth parameters mentioned and develop an intelligent control action clas-
211 sification system from the sensor data obtained while also implementing a
212 method to make the system much more resilient to erroneous sensor data.

213 3. Proposed System

214 A traditional NFT hydroponics system comprises mainly two sections:
215 the growth channels, which are placed at an elevation, and the reservoir below
216 it[17]. An electric pump moves nutrient solution into the growth channels,
217 where the plant roots come in contact with the solution. The solution then
218 moves down the slope of the channels and back to the reservoir. For the
219 system we designed, we split it into two sections:

- 220 1. **Growth Module (GM)**: This is where the plants are housed and
221 grow from seedlings to the end of maturation.
- 222 2. **Nutrient Solution Container Module (NSCM)**: This is where the
223 nutrient solution given to the plants is primarily stored while constantly
224 recirculating.

225 The two modules are connected via tubing and pipes to ensure re-circulation
226 and nutrient recycling [25].

227 3.1. Growth Module

228 This is the module where the plants are grown. The mechanism with
229 which our proposed GM works is as follows:

- 230 1. An air temperature and humidity sensor (DHT22 Temperature and
231 Humidity sensor) measures the module's ambient air temperature and
232 humidity. A thermoelectric cooler regulates the temperature and keeps
233 it within the desired range.
- 234 2. A Carbon Dioxide (CO₂) sensor (MH-Z19C Nondispersive Infrared
235 (NDIR) CO₂ sensor module) monitors the concentration of CO₂ gas
236 within the system. If the concentration of CO₂ gas falls below the op-
237 timal range, a normally closed pneumatic solenoid valve, attached to
238 a cylinder containing CO₂ gas, is be actuated to allow the inflow of
239 CO₂ gas into the system to bring the concentration within the optimal
240 range.
- 241 3. A camera (Lenovo Select FHD Webcam) is used to provide a real-time
242 video feed of the interior of the Growth Module. This allows us to
243 monitor the plants' physical state during the growth cycle.

244 4. LED arrays (two Auoyo 40W LED Growlight lamps) provide a con-
245 sistent and adequate alternative to sunlight for the plants. Since we
246 tested the system by growing lettuce, the LEDs are kept on for 16
247 hours, followed by 8 hours of complete darkness by turning them off,
248 which provides the optimal daylight hours for their growth [26].

249 3.2. Nutrient Solution Container Module

250 This module will hold the nutrient solution to be used in our system.
251 The nutrient solution will mainly constitute Nitrogen(N), Phosphorus(P),
252 and Potassium(K), with the addition of a multitude of other macronutrients
253 and micronutrients. The mechanism with which our proposed NSCM works
254 is as follows:

- 255 1. A pH sensor (Atlas Scientific Analog pH Kit) constantly measures the
256 pH level of the nutrient solution in the container. A dosing system
257 is designed to deliver acid or base to the module using pumps and
258 tubing. If the pH level is outside the optimal pH range required for
259 the specific plant type, the dosing system adds acid or base to the
260 nutrient solution to bring the pH back within the optimal range. After
261 adding a set amount of acid or base, the dosing system is disabled
262 for 10 minutes. This ensures that the added acid or base has ample
263 time to mix thoroughly within the nutrient solution to bring about the
264 intended pH change before adding an additional amount.
- 265 2. A waterproof ultrasonic distance sensor (JSN SR-04T Waterproof Ul-
266 trasonic Sensor) measures the height at which the nutrient solution is
267 present to determine the amount of nutrient solution within the sys-
268 tem. If the height of the nutrient solution falls or exceeds drastically,
269 the central server will issue a warning to the user.
- 270 3. An EC sensor (Atlas Scientific Mini Conductivity K 1.0 Kit) measures
271 the concentration of nutrient ions in the solution if the EC reading is
272 outside the optimal EC range required for the specific plant type. In
273 that case, the dosing system adds a concentrated nutrient solution or
274 distilled water to the nutrient solution to bring the EC readings back
275 within the optimal range. Similar to when adding acid or base, the
276 dosing system is disabled for the next 10 minutes. This ensures that
277 the added distilled water or concentrated nutrient solution has ample

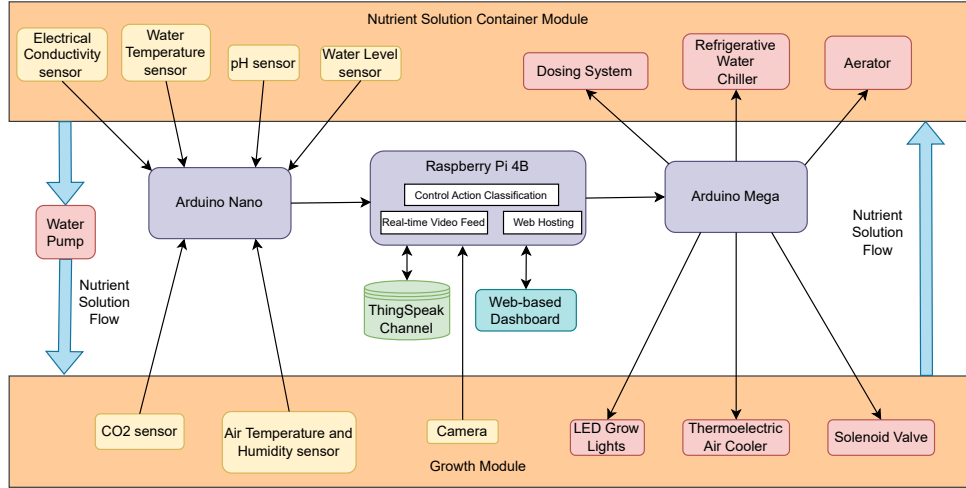


Figure 1: Overview of the proposed system illustrating how the various sensors and actuators are configured with the microcontrollers and Raspberry Pi 4B

- time to mix completely within the nutrient solution to bring about the intended change in EC before an additional amount is added.
4. A refrigerative water chiller (Hailea HS-28A Aquarium Chiller) regulates the temperature of the nutrient solution. The heat from the external environment and the LED grow lights cause the temperature of the nutrient solution to go up. The refrigerative water chiller efficiently reduces the temperature and maintains it within the optimal range.
 5. An aerator oxygenates the nutrient solution and helps the dissolved nutrients mix evenly. Since the nutrient solution is required to maintain a minimum Dissolved Oxygen (DO) level of 7PPM [27], it is kept running throughout the growth cycle to ensure that the DO remains above the required level.
 6. A water pump constantly pumps the nutrient solution to the GMs as it returns to the NSCM, creating a non-stop flow of nutrient solution, ensuring that the plant roots receive a constant supply of water, dissolved oxygen, and nutrients.



Figure 2: The system running and operational before the first batch trial

3.3. Control Unit

The sensors used in the GM and NSCM are interfaced with an Arduino Nano, while the actuators are interfaced with an Arduino Mega. Both microcontrollers were connected to a Raspberry Pi 4 Model B, which served as the central control unit for the system. Sensor readings from the sensor were processed using median filters in the Arduino Nano before being sent to the Raspberry Pi 4, which simultaneously updated the sensor data to a ThingSpeak channel and determined the appropriate control actions. The control actions were passed onto the Arduino Mega, which activated the actuators to regulate the physical conditions within the system. Readings were taken at 15-second intervals. An illustration of the connections between the various components of the proposed system can be seen in Figure 1. It shows how each part of the proposed system communicates to ensure optimal conditions are kept within it while allowing users to monitor said physical parameters.

A web-based dashboard was developed and hosted on the Raspberry Pi 4, with data visualizations and widgets provided by ThingSpeak to monitor the physical conditions within the system. A real-time video feed from within the GM was also embedded in the dashboard so that the physical well-being of the lettuce could be monitored visually. Figure 2 shows the entire system before it was tested with lettuce for a growth period of 30 days.

315 4. Experimentation with growing the first batch

316 The first experiment was designed as an initial test for the system against
317 traditional farming techniques. 42 Green Leaf lettuce seedlings, grown from
318 locally procured seeds, were randomly divided into three equal batches of 14
319 seedlings. Each of these batches was then transplanted into the following
320 three treatments on the same day:

- 321 1. P1: Our proposed system
- 322 2. P2: 50% Soil + 50% vegetable compost in pots
- 323 3. P3: Manual Deep Water Culture (DWC) hydroponics system

324 The experiments were conducted in the Dhaka metropolitan area, Bangladesh
325 (23.7445° N latitude and 90.3654° E longitude) during December and Jan-
326 uary. Throughout the 30-day growth period, the ambient temperatures
327 ranged between 15°C and 26°C . Lettuce plants in P2 and P3 were kept
328 close at the exact location to ensure equal exposure to sunlight, tempera-
329 ture, humidity, and ambient CO_2 concentration. The nutrient solution used
330 in P1 and P2 was prepared using the concentrated solution of Nutrient A
331 and B, ensuring that the initial EC value of both solutions is kept as close as
332 possible. Data was collected daily, starting the day after the Day of Trans-
333 plant (DAT). The plants were grown in all three treatments for 30 days, and
334 throughout this duration, the plants were photographed from a top view.
335 The sensor data from our proposed system was stored on the ThingSpeak
336 channel, using which we developed our intelligent classification method and
337 the algorithm for error resilience.

338 4.1. *Determining Erroneous Sensor Data and Developing the Error Resilience* 339 *Algorithm*

340 A combination of median and bandpass filters was applied to the sensor
341 data received by the Arduino Nano before being passed onto the Raspberry
342 Pi. The optimal range for values for the physical parameters regulated by
343 the proposed system is below. Considering the proposed system does not
344 actively regulate humidity, we left it out of this list:

- 345 (i) pH: Between 5.8 and 6.5
- 346 (ii) Electrical Conductivity: Between 1,000 and 1,500 microSiemens cm^{-1}

- 347 (iii) Water Temperature: Between 18 and 22°C
- 348 (iv) Air Temperature: Between 20 and 26°C
- 349 (v) CO₂ Concentration: Between 1,900 and 2,100 PPM
- 350 (vi) Water Level: Between 4 and 15 cm

351 Despite the use of filters, instances of erroneous data were abundant.
352 Inexplicable spikes in sensor data were common and expected, some spon-
353 taneous, which stabilized almost immediately. However, there are also nu-
354 merous instances when the readings would keep spiking until they reached
355 values not physically possible within the system. The pH sensor often logged
356 increasingly high pH values, going up to 10 or higher, even when the dosing
357 system was not adding any base to the nutrient solution. This led to the
358 dosing system adding acid into the nutrient solution, sharply decreasing its
359 pH. Situations like these had to be dealt with promptly to ensure the plants
360 were not irreversibly damaged.

361 We determined the instances of erroneous data by going through the
362 sensor data exported from ThingSpeak. The arbitrary criterion for being an
363 erroneous data point is that the value obtained exceeds the lower or upper
364 limits of the optimal range by 10%, which also considers a flat 5% sensing
365 uncertainty. Furthermore, we had to assume that the actuators within the
366 system worked optimally at all times. Table 1 shows both the cumulative
367 instances of errors as well as for each sensor. A total of 53,834 instances
368 of erroneous data were found against 742,458 cumulative instances of sensor
369 data (adding instances for values being taken for each sensor), which accounts
370 for 7.25% of all the data points received by the Raspberry Pi.

Sensor	Instances of Errors
EC	1162 (0.16%)
pH	1802 (0.24%)
Air Temperature	0 (0.00%)
Water Temperature	1849 (0.25%)
Water Level	43761 (5.89%)
CO ₂ Concentration	5260 (0.71%)
Total Instances of Errors	53834 (7.25%)

Table 1: Instances of erroneous data from sensors received by the Raspberry Pi during the first experiment

We could not identify a single concrete reason for these erroneous data. However, we suspect leakage current in the circuits and nutrient solution, the latter due to the presence of a submersible water pump, to play a part in them. The accuracy of the EC, pH, and water temperature sensors was paramount among all the sensor readings. Since even the slightest unnecessary addition of acid or base to the nutrient solution would almost immediately cause the pH of the entire solution to go beyond the optimal range. Especially during the early stages, plant roots are highly sensitive to pH, especially high and low pH conditions [28], [29]. Similarly, a high value of EC is likely to damage roots [30]. Furthermore, the very high or very low water temperature will inhibit root hair growth and consequently plant growth itself [31], [32]. When passed to the control action classification function, these erroneous sensor data led the system to perform unnecessary actions, which caused unnecessary changes to the growth parameters. In some instances, the pH of the nutrient solution changes abruptly and by large margins, which had to be fixed manually immediately to prevent any damage to the plants. To prevent this, we designed an algorithm to ensure that even with instances of erroneous data, the system does not act on them.

Here is how the algorithm works with a modified combination of threshold filters and bandpass filters, being implemented in two steps:

1. Within the sensing loop: Let us consider a sensor in the system, S , which measures the growth parameter, P .
 - (a) At the initial startup, the Arduino Nano processes the signal received from the sensor by applying a combination of median and

395 bandpass filters. After a warm-up period of 5 minutes for the
 396 sensors, the filtered signal value is assigned to the variable V_T .
 397 (b) A subsequent filtered signal value is received from the Arduino
 398 Nano and is assigned to the variable V_n . Following which, it is
 399 checked if the sensing reliability criterion is met, given by:

$$|V_n - V_T| \leq (1 + \beta)\xi$$

400 (c) ξ is the error tolerance threshold, which is determined based on
 401 the distribution of sensor data for the parameter S , and β is an
 402 arbitrary error rounding multiplier such that $0 \leq \beta \leq 1$. Let's
 403 consider that for sensor, S , the number of stored sensor data, S_V ,
 404 is M . To that extent, S_V^i denotes the i^{th} stored sensor value. ξ is
 405 calculated as such:

$$\xi = \sum_{i=1}^{M-1} \frac{|S_V^i - S_V^{i+1}|}{M}$$

406 This calculates the mean change in sensor values, which helps to
 407 determine the typical change in sensor readings and reduces the
 408 impact of outliers if the value of m is sufficiently large.

409 (d) If the sensing reliability criterion is satisfied, then V_T is updated
 410 with the value of V_n . The updated value V_n is subsequently passed
 411 to the actuation loop. If the reliability criterion is not met, the
 412 V_n value is discarded, and the sensing loop restarts.
 413 (e) As the sensing reliability criterion is checked, another counter, ω ,
 414 is initialized. It is incremented for every consecutive instance of
 415 V_n that does not satisfy the sensing reliability criterion and is reset
 416 to 0 whenever a value of V_n satisfies it. This is expressed by:

$$\omega = \sum_{i=1}^M \mathbb{I}(|V_{n_i} - V_{T_i}| > (1 + \beta)\xi)$$

417 Where, $\mathbb{I}(\cdot)$ is an indicator function that results in a 1 if the condi-
 418 tion inside the function, in this case, the sensing reliability crite-
 419 rion, is met. Otherwise, it outputs a 0. We designed this indicator

function to output 1 whenever V_n fails to satisfy the sensing reliability criterion. Following this, while ω is incremented, the sensing loop reset criterion, given by $\omega \geq 50$, is checked. The threshold value of 50 was chosen as an arbitrary value as it seemed like a reasonable number of values that should be evaluated to check the sensing reliability criterion.

- (f) Once this criterion is satisfied, the sensing loop resets with the current value of V_n being assigned to V_T . It should be noted that in cases where the sensor is constantly registering erroneous data, silently resetting the sensing loop might lead to the passing of inaccurate sensor data to the actuation loop. When designing the actuation loop, we considered this and implemented a notification alert using the PushBullet API to send messages to the user whenever the sensing loop is reset. This prompts the user to check the web-based dashboard to see whether the sensor functions as it should.

2. Within the actuation loop:

- (a) For the optimal range of the growth parameter, P , instead of setting discrete upper and lower limit values, we define actuation bands on both ends of the range.
- (b) Considering the case for sensor, S , measuring parameter, P , the optimal range is between L_{lower} and L_{upper} . Instead of setting these fixed values, we determined a range in which the required actuation is undertaken if the sensor value happens to fall in.
- (c) The width of the actuation bands, ζ , is dependent on the upper limit, L_{upper} , the lower limit, L_{lower} , an arbitrary threshold multiplier, α , and ξ . The actuation bands are defined as:

$$\zeta_{upper}(L, \alpha, \xi) \in [L_{upper} - \alpha\xi, L_{upper} + \alpha\xi]$$

$$\zeta_{lower}(L, \alpha, \xi) \in [L_{lower} - \alpha\xi, L_{lower} + \alpha\xi]$$

- (d) If V_n is within ζ , a counter, μ_k , is initialized and counts the number of successive instances for which V_n falls within ζ . This is formalized as:

$$\mu_k = \begin{cases} 0 & \text{if } V_n \notin [\zeta_{\text{lower}}, \zeta_{\text{upper}}] \\ \mu_{k-1} + 1 & \text{if } V_n \in [\zeta_{\text{lower}}, \zeta_{\text{upper}}] \end{cases}$$

451 (e) From the sensor data collected, the mean number of consecutive
 452 instances where the values are within 2ξ is calculated. This is the
 453 repeatability index, denoted by ψ .

$$\psi = \left\lfloor \frac{1}{k} \sum_{j=1}^k \text{length}(\eta_j) \right\rfloor$$

454 where, k is the number of sequences η_j of consecutive instances
 455 such that $|S_V^i - S_V^{i+1}| \leq 2\xi$ for $i = 1, 2, 3, \dots, M$ in the stored
 456 sensor values, S_V . ψ is rounded down to the nearest integer since
 457 it represents instances of a count.

458 (f) If the sensing repeatability criterion is met, given by:

$$\mu_k \geq \psi$$

459 V_n is passed onto the classification method. The control action
 460 is determined, and the actuation commands are passed onto the
 461 appropriate actuator. The actuation loop then resets. If successive
 462 values of V_n exceed the bounds of ζ before reaching the count of
 463 ψ , the value of μ_k is reset to 0.

464 The values of multipliers α and β are different for each sensor, owing to
 465 their varying sensitivity, stability, and overall quality. As such, for each sen-
 466 sor, S , values of α_S and β_S have to be determined via simultaneous iteration.
 467 It must also be noted that the values of α_S and β_S , if large, will allow a
 468 greater number of sensor values through to the classification method since
 469 the difference between V_T and V_n will be allowed to be larger. At the same
 470 time, the actuation bands will also bolster a larger width from ζ . This is not
 471 ideal since we want the algorithm to pass sensor values that are valid and
 472 devoid of errors. As such, the values of α_S and β_S need to be kept minimal.
 473 Hence, the algorithm is treated as a multivariate optimization problem with
 474 constraints, where the objective is to minimize the instances of errors for a
 475 sensor, E_S . The criterion for a sensor value to be counted in E_S is the same as
 476 in the first experiment: values exceeding the 10% tolerance threshold beyond

477 the lower or upper limits of the optimal range. In this case, the criterion is
 478 altered to consider the upper limit of ζ_{upper} and the lower limit of ζ_{lower} . As
 479 such, we introduce Lagrange multipliers, λ_{1_S} and λ_{2_S} , to allow for efficient
 480 optimization for our arbitrary multipliers. It can be formalized as such:

$$\begin{aligned} & \underset{\alpha_S, \beta_S}{\text{Minimize}} && E_S(\alpha_S, \beta_S) + \lambda_{1_S}\alpha_S + \lambda_{2_S}\beta_S \\ & \text{subject to} && 0 < \alpha_S \leq 1 \\ & && 0 \leq \beta_S \leq 1 \end{aligned}$$

481 We take the approach of minimizing the errors for each sensor rather than
 482 minimizing the sum of errors of all sensors combined to ensure that there are
 483 minimal erroneous actuations corresponding to each parameter, P , resulting
 484 from erroneous sensor readings.

485 To make the computation of the optimization more straightforward, we
 486 propose a cost function, C_S , such that:

$$C_{S_{i,j}} = \lambda_E \frac{E_{S_{i,j}}}{E_{S_{\max}}} + \lambda_\alpha \frac{\alpha_{S_i}}{\alpha_{S_{\max}}} + \lambda_\beta \frac{\beta_{S_j}}{\beta_{S_{\max}}}$$

487 Where:

- 488 • $E_{S_{i,j}}$ is the total error count for the α_{S_i} and β_{S_j} combination.
- 489 • α_{S_i} is the i -th iterative value of α_S .
- 490 • β_{S_j} is the j -th iterative value of β_S .
- 491 • $E_{S_{\max}}$ is the maximum observed error count for sensor S .
- 492 • $\alpha_{S_{\max}}$ is the maximum value of alpha.
- 493 • $\beta_{S_{\max}}$ is the maximum value of beta.
- 494 • λ_E is the weight for the error term.
- 495 • λ_α is the weight for the alpha term.
- 496 • λ_β is the weight for the beta term.

497 To determine optimal values of α_S and β_S , we proposed an adaptive multi-
 498 parameter λ -weighted minimization approach, similar to gradient descent
 499 approaches [33] conventionally used in machine learning, with the following
 500 update policies, given by π .

501 Considering for the sensor, S :

$$\begin{aligned}\pi_\alpha : \alpha_i &\leftarrow \alpha_{i-1} \left[1 - \gamma \left(\log \left(\frac{C_{S_{i-1},j-1}}{C_{S_{i,j}}} \right) + \frac{\gamma}{i^2} \right) \right] \\ \pi_\beta : \beta_j &\leftarrow \beta_{j-1} \left[1 - \gamma \left(\log \left(\frac{C_{S_{i-1},j-1}}{C_{S_{i,j}}} \right) + \frac{\gamma}{j^2} \right) \right] \\ \pi_{\lambda_\alpha} : \lambda_\alpha &\leftarrow \lambda_\alpha \left[1 - \log \left(\frac{C_{S_{i-1},j-1}}{C_{S_{i,j}}} \right) \right] \\ \pi_{\lambda_\beta} : \lambda_\beta &\leftarrow \lambda_\beta \left[1 - \log \left(\frac{C_{S_{i-1},j-1}}{C_{S_{i,j}}} \right) \right] \\ \pi_{\lambda_E} : \lambda_E &\leftarrow \lambda_E \left[1 + \log \left(\frac{C_{S_{i-1},j-1}}{C_{S_{i,j}}} \right) \right]\end{aligned}$$

502 The proposed optimization framework employs a set of adaptive update
 503 policies designed to adjust multiple parameters dynamically throughout the
 504 iterative process. These policies are formulated to refine the search space
 505 progressively and shift the optimization focus as the algorithm converges.

506 The update policies for α and β are constructed to ensure a monotonic
 507 change in their respective values. If the value of the cost function is reduced,
 508 the value of α and β decreases. Inversely, an increase in cost function value
 509 increases their respective values, indicating that the smaller values of α and
 510 β were not closer to the optimal values but farther away. The gradual re-
 511 duction in α and β is crucial for the optimization process, as it allows for
 512 more significant parameter adjustments in the early stages when the algo-
 513 rithm is likely far from the optimal solution. As the optimization progresses,
 514 the decreasing values of α and β lead to smaller, more refined adjustments,
 515 facilitating convergence to the optimal solution. The inclusion of the $\frac{\gamma}{i^2}$ and
 516 $\frac{\gamma}{j^2}$ terms, where γ represents the learning rate, serves a crucial role in the
 517 early stages of optimization. Initially, the logarithmic term may be close to
 518 zero, providing little guidance for updates. The $\frac{\gamma}{i^2}$ and $\frac{\gamma}{j^2}$ terms compensate
 519 for this by adding a more significant effect in the initial iterations. As the
 520 number of iterations, i , and j , increase, the influence of the terms diminishes,

eventually becoming negligible. This design allows for larger updates early on when the algorithm is likely far from the optimal solution and smaller, more refined updates as it approaches convergence.

To avoid convergence to local minima, the algorithm employs multiple random initializations for α and β . Specifically, the process is repeated ten times with different initial values, allowing the optimization to explore various starting points in the parameter space. This approach increases the likelihood of finding the global optimum by sampling a broader range of initial conditions. The λ parameters introduce a temporal dynamic to the optimization process. The gradual reduction in λ_α and λ_β diminishes the influence of α and β on the cost function over time, while the increasing λ_E amplifies the weight of the error term. This mechanism facilitates a transition from broad parameter space exploration to focused error minimization as the optimization progresses.

The synergy between these update policies creates an adaptive optimization framework that efficiently navigates the parameter space. By dynamically adjusting the balance between exploration and exploitation, this approach aims to mitigate the risk of convergence to local optima while promoting robust global optimization performance. The combination of decreasing α and β values, multiple random initializations, and the evolving influence of λ parameters allows the algorithm to thoroughly explore the solution space initially and then gradually focus on refining the most promising solutions.

C_S combines E_S , α_S and β_S values for deviations in sensor values, each weighted according to their importance. This approach ensures a comprehensive and quantifiable measure of sensor error minimization, allowing the computational approach to quickly settle on the best values for the multipliers that result in minimal instances of errors. $C_{S_{ij}}$ considers normalized components, making it easier to offset the difference in individual scaling for each component. The initial values of α_S and β_S , we set the values of λ_E , λ_α and λ_β are all set to 1.0. The sum of C_S for all the sensors, referred to as $\sum C_S$, is also calculated, which acts as an aggregate benchmark to show the extent to which instances of erroneous sensor data are mitigated.

From the sensor data collected from the first experiment, we fed the data into the algorithm to determine the optimal values of α_S and β_S for each sensor by minimizing $C_{S_{ij}}$. For 10 random initializations of α_S and β_S , the values of γ were set at 0.25, 0.5, and 0.75, treating it as a hyper-tunable parameter. Table 2 shows the values of α_S and β_S determined from the optimisation. We observed that $\gamma = 0.75$ yielded the lowest total cost

Sensor Data	Multiplier Values					
	$\gamma = 0.25$		$\gamma = 0.5$		$\gamma = 0.75$	
	α	β	α	β	α	β
EC	0.0557	0.0579	0.0045	0.0026	0.0006	0.0002
pH	0.0516	0.0599	0.0406	0.0128	0.0007	0.0005
Air temperature	0.0232	0.1597	0.0078	0.0125	0.0002	0.0022
Water temperature	0.2074	0.0654	0.0127	0.0066	0.0006	0.0009
Water level	0.1057	0.0793	0.0140	0.0251	0.0001	0.0017
CO ₂ concentration	0.0881	0.0420	0.0172	0.0119	0.0003	0.0010
$\sum C_S$	1.6000		0.7726		0.6133	

Table 2: Optimal α_S and β_S values for each sensor measuring the given physical parameters for $\gamma = 0.25, 0.5$ and 0.75

function value of 0.6133 from these sets. We settled on these α_S and β_S values, which are later used in the error resilience algorithm implemented in the second experiment while growing the second batch of lettuce.

4.2. Training Machine Learning Models for Control Action Classification

The collected sensor data had to be cleaned of anomalous values and rows with missing data caused by unknown technical disturbances. We then performed data augmentation to make the dataset more balanced, prioritizing classes with the lowest instances. This left us with 110,800 data points. These data points are used to construct the dataset $D = \{(x_i, y_i), \quad i = 1, 2, \dots, N\}$, such that

$$x_i = [x_{i1}, x_{i2}, x_{i3}, \dots, x_{ip}]$$

Is a row vector that contains the input variables, i.e., sensor data (or features) and

$$y_i \in \{0, 1, 2, \dots, M\}$$

Is the scalar integer output. In our case, output is the control actions determined by the Raspberry Pi with respect to the sensor data. These actions are either ON or OFF for the given actuators:

- 574 1. Acid Doser pump
- 575 2. Base Doser pump
- 576 3. Nutrient Doser pump
- 577 4. Distilled Water Doser pump
- 578 5. Air Chiller
- 579 6. Water Chiller along with its water pump
- 580 7. CO₂ Delivery solenoid

581 Considering the two possible action states, ON as 1 and OFF as 0, the
 582 number of possible control action combinations determined by the Raspberry
 583 Pi is $2^7 = 128$.

584 Adapting the One-Hot encoding approach, the control action values of 1s
 585 and 0s are added to each row in the CSV dataset. The seven encoded cate-
 586 gorical columns for the control actions are later appended into a 7-bit binary
 587 number in the 'Binary_Label' column of the dataset, which in turn is con-
 588 verted into its corresponding decimal value and stored in the 'Decimal_Label'
 589 column. These values are the labels, y_i , for the model. The dataset was then
 590 used to train machine learning models, namely Logistic Regression, Decision
 591 Tree, Random Forest, Support Vector Machine(SVM), K-Nearest Neighbour
 592 (KNN), Gradient Boosting and XGBoost. The features used to train the
 593 models were the physical growth parameters that were monitored and con-
 594 trolled within the system.

Models	Performance Metrics			
	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.942	0.827	0.759	0.771
Decision Tree	1.000	1.000	0.999	0.999
Random Forest	1.000	0.999	0.992	0.996
SVM	0.911	0.706	0.701	0.686
KNN	0.936	0.878	0.859	0.858
Gradient Boosting	0.776	0.488	0.463	0.426
XGBoostClassifier	1.000	1.000	1.000	0.999

Table 3: Performance metrics of different machine learning models

From Table 3, we see that the XGBoostClassifier performs the best with scores of 1.000 for accuracy, precision, and recall. The F-1 score is also joint highest with Decision Tree at 0.999. We opted to go with the XGBoostClassifier as it builds more robustly on the principle of Decision Trees. The model trained with this data was later used for real-time inferencing for control action classification for the second batch of lettuce.

5. Experimentation with Growing the second batch

The second experiment was mainly designed to test the efficacy of our system at higher ambient temperatures, at which lettuce struggles to grow using traditional soil-based agricultural methods. The error resilience algorithm and the XGBoostClassifier model for the intelligent classification method were also evaluated in this experiment. For this setup, 21 Green Leaf lettuce seedlings, grown from the same locally procured seeds as in the first experiment, were randomly divided into three equal batches of 7 seedlings. Each of these batches was then transplanted into the three treatments, P1, P2, and P3, on the same day.

However, unlike the first experiment, which was done during winter, the second experiment was done during March and April, which is typically spring or the start of summer in this region. The intention of growing the second batch of lettuce during this time was to test the efficacy of our system during an off-season for the lettuce, which typically grows the best when the temperatures are lower [34]. The ambient temperature during this duration was between 25°C and 36°C. The location of the experiments was kept the same throughout a similar 30-day growth period to ensure that P2 and P3 had similar ambient conditions as in the first experiment. The sensor data from the second experiment were also stored on the ThingSpeak channel. This data was used to test the efficacy of the intelligent classification method and error resilience algorithm.

After the 30-day growth period for both experiments, the lettuce plants from the three treatments were harvested. For lettuce harvested from both experiments, their agronomic parameters were taken, which include Fresh and Dry Mass of Aerial Part (FMAP and DMAP), Total Plant Length (TPL), Root Length (RL), Shoot Length (SL), Total Number of Leaves (TNL) and Number of Leaves longer than 10cm (NL10). FMAP is the weight of the plant's above-ground parts immediately after harvesting, reflecting water content and biomass, while DMAP is the weight after drying, indicating

631 structural biomass. TPL is the combined length of the root and shoot sys-
632 tems, measured to assess overall vertical growth. RL and SL are the lengths
633 of the root and shoot systems, respectively, measured from base to tip to
634 evaluate growth. TNL is a count of all leaves on the plant, indicating veg-
635 etative growth, and NL10 counts only leaves longer than 10cm, providing
636 information on the development of larger leaves. The lettuce plants were
637 also tested for nutritional data, namely the content of macronutrients such
638 as protein, fat, and carbohydrates.

Agronomic Parameters	Treatment					
	First Experiment			Second Experiment		
	P1	P2	P3	P1	P2	P3
FMAP (g)	252.87	141.98	94.09	264.81	68.24	109.49
DMAP (g)	34.82	16.56	12.38	33.19	8.63	13.85
TPL (cm)	81.01	57.27	20.67	72.52	19.46 ^a	20.12 ^a
RL (cm)	35.68	27.13	6.09	25.33	8.48 ^a	9.31 ^a
SL (cm)	45.33 ^a	30.14 ^a	14.58	47.63	10.98 ^a	10.80 ^a
TNL (cm)	20.29	14.21	11.93	20.29	9.86 ^a	11.86 ^a
NL10 (cm)	10.21 ^a	9.5 ^a	7.07	13.00	3.00	7.57

All averages across all the rows differ at the 0.1% significance level for the one-way ANOVA.

^a No statistically significant differences were observed among the mean values across rows from the Games-Howell post-hoc test at the 5% significance level, following the one-way ANOVA.

Table 4: Average values for agronomic parameters: Fresh and Dry Mass of Aerial Part (FMAP and DMAP), Total Plant Length (TPL), Root Length (RL), Shoot Length (SL), Total Number of Leaves (TNL) and Number of Leaves longer than 10cm (NL10) from the three treatments from both experiments

639 6. Results

640 Table 4 compares the agronomic parameters from the lettuce grown in
641 both experiments in all three treatments. In the first experiment, treatment

P1 consistently exhibits the highest values across most agronomic parameters. Specifically, P1 shows the greatest Fresh and Dry Mass of the Aerial Part (FMAP: 252.87 g; DMAP: 34.82 g), significantly surpassing both P2 and P3, with P1's FMAP being 78.1% higher than P2 and 168.8% higher than P3. For Total Plant Length (TPL), P1 again leads with the longest length (81.01 cm), notably longer than P2 and P3; P1's TPL is 41.5% longer than P2 and 291.9% longer than P3. P1 also has the longest Root Length (RL: 35.68 cm), significantly greater than P2 and P3, with P1's RL being 31.5% greater than P2 and 485.9% larger than P3. Shoot Length (SL) follows a similar trend, with P1 showing the greatest length (45.33 cm) compared to P2 and P3. Regarding the Total Number of Leaves (TNL), P1 has the highest count (20.29), though the differences between treatments are less pronounced here. For the Number of Leaves Longer Than 10 cm (NL10), P1 leads with 10.21, higher than P2 and P3. In the second experiment, P1 demonstrates the highest FMAP (264.81 g) and DMAP (33.19 g), showing that P1's FMAP is 288.1% higher than P2 and 141.9% higher than P3 in the second experiment. but the differences between treatments are less stark compared to the first experiment.

While P1 maintains the highest TPL (72.52 cm), the differences between treatments are smaller. P1 also has the longest RL (25.33 cm), but the differences are less pronounced than in the first experiment. For SL, P1's value (47.63 cm) is notably higher than P2 and P3, which are pretty similar. TNL remains stable across experiments for P1 (20.29), with P2 and P3 showing minor differences. Finally, P1 exhibits a significant increase in NL10 (13.00) compared to P2 (3.00) and P3 (7.57), indicative of P1's NL10 being 333.3% higher than P2 and 71.7% higher than P3. This highlights a more significant disparity between treatments in the second experiment.

Adding to that, the observations from Figure 3 show that while treatment P1 shows superior performance across most parameters in both experiments, the second experiment reveals more minor differences between treatments, indicating that the change in ambient temperature during the time of the second experiment had a considerable effect on the growth of the plant. Most notably, P1 consistently demonstrates superior performance across most parameters in both experiments. This superiority is particularly evident in metrics such as Fresh Mass of Aerial Part (FMAP), Total Plant Length (TPL), and Number of Leaves longer than 10cm (NL10), suggesting that P1 has a significant positive impact on overall plant growth and development.

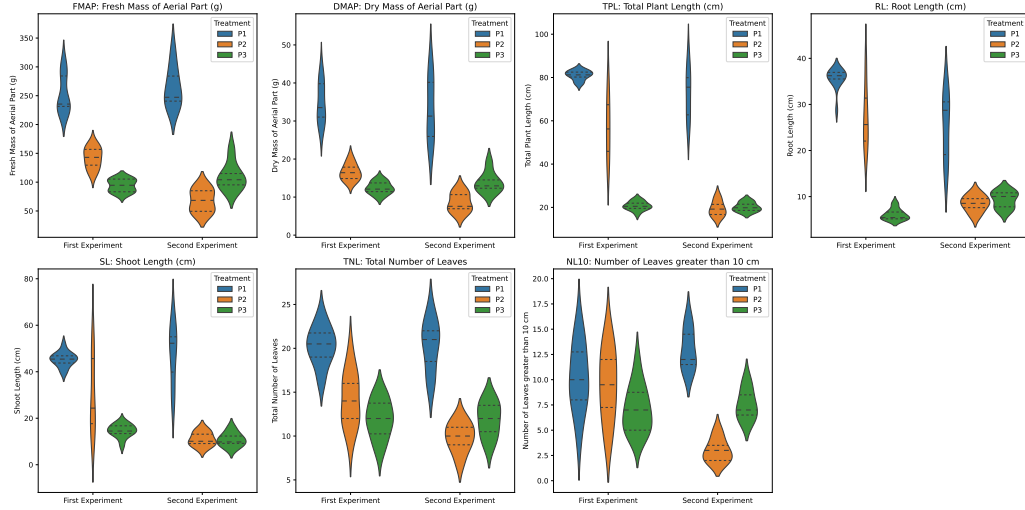


Figure 3: Violinplots illustrating the comparison between the agronomic parameters for the lettuce grown in the three treatments: P1, P2 and P3 across both experiments. Fresh and Dry Mass of Aerial Part (FMAP and DMAP), Total Plant Length (TPL), Root Length (RL), Shoot Length (SL), Total Number of Leaves (TNL) and Number of Leaves longer than 10cm (NL10)

680 Interestingly, while the second experiment generally shows lower values
681 and reduced variability compared to the first, the relative performance of
682 treatments remains consistent. The data also reveals that above-ground
683 growth parameters (e.g., Shoot Length, Total Number of Leaves) show more
684 pronounced treatment effects than below-ground parameters like Root Length.
685 The Fresh and Dry Mass of Aerial Part (FMAP and DMAP) exhibit the
686 widest range of values, indicating high treatment sensitivity. Notably, P1
687 often displays a wider distribution in the upper range for these and other
688 parameters, suggesting a potential for higher maximum values and, by ex-
689 tension, higher yield potential. In contrast, P2 and P3 generally show more
690 compact distributions, indicating more consistency but lower overall perfor-
691 mance. Overall, the comparison between the control groups and the lettuce
692 grown in our proposed system is markedly better across the different weather
693 conditions in both experiments. Considering nutritional data, Table 5 high-
694 lights how in the first experiment, P3 had the highest protein content at 0.42
695 $\text{g}(100 \text{ g})^{-1}$, compared to P1 at 0.39 and P2 at 0.35. This indicates that P3
696 was the richest in protein among the three treatments; P3's protein content
697 was 7.7% higher than P1 and 20% higher than P2. For fat content, P1 and

P3 had the lowest fat levels at 0.07, while P2 had a slightly higher fat content of 0.09. Regarding Total Dietary Fiber (TDF), P3 had the highest fiber content at 0.96, followed by P2 at 0.61 and P1 at 0.53. Regarding carbohydrates, P2 contained the highest amount at 3.28, compared to P1 at 1.72 and P3 at 1.24.

In the second experiment, P2 had the highest fat content at $4.52 \text{ g}(100 \text{ g})^{-1}$, significantly higher than P1 at 2.20 and P3 at 1.99. For protein, P1 led with 5.39, followed by P3 at 4.78 and P2 at 3.18, indicating that P1 had the highest protein content in the second experiment; P1's protein content was 69.5% higher than P2 and 12.8% higher than P3. The TDF content was highest in P2 at 1.34, compared to P1 at 1.15 and P3 at 1.27, showing that P2 had the greatest dietary fibre. Lastly, P3 had the highest carbohydrate content at 3.83, with P2 at 2.83 and P1 at 2.14.

Macronutrient ($\text{g}(100 \text{ g})^{-1}$)	Treatment					
	First Experiment			Second Experiment		
	P1	P2	P3	P1	P3	P3
Protein	0.39	0.35	0.42	5.39	3.18	4.78
Fat	0.07	0.09	0.07	2.20	4.52	1.99
Total Dietary Fiber (TDF)	0.53	0.61	0.96	1.15	1.34	1.27
Carbohydrates	1.72	3.28	1.24	2.14	2.83	3.83

Table 5: Average macronutrient content from the first and second experiments. The values represent the content of Protein, Fat, Total Dietary Fiber (TDF), and Carbohydrates in grams per 100 grams ($\text{g}(100 \text{ g})^{-1}$) and are from across three treatments (P1, P2, and P3) in both experiments.

Overall, differences between the treatments within each experiment reveal that P3 was the most protein-rich and fiber-rich in the first experiment, while P2 had the highest fat and carbohydrate content. In the second experiment, P1 showed the highest protein content, P2 had the greatest TDF, and P3 had the highest carbohydrate level. These findings highlight how the change in ambient conditions affected the nutrient profile of plants in P2 and P3 the most due to their exposure to said conditions.

As for the error resilience algorithm, Table 6 illustrates the decrease in instances of erroneous sensor data. Compared to the first experiment, there were 1.84% instances of errors, which is ≈ 3.94 times lower than the 7.25% of the first. This highlights the overall efficacy of the algorithm, which

Sensor	Instances of Errors
EC	0 (0.00%)
pH	2725 (0.37%)
Air Temperature	0 (0.00%)
Water Temperature	0 (0.00%)
Water Level	0 (0.00%)
CO ₂ Concentration	10706 (1.47%)
Total Instances of Errors	13431 (1.84%)

Table 6: Instances of erroneous data from sensors received by the Raspberry Pi during the second experiment

significantly reduces the chances of passing erroneous data onto the classification method. As for the intelligent classification method itself, the XGBoostClassifier-based intelligent classification method was able to determine control actions correctly almost all the time, with an accuracy score of 0.98, with recall, precision, and F1-score of 0.97, 0.96, and 0.96 respectively. This shows that the machine learning-based approach to control action determination holds its ground against conventional if-else methods.

7. Conclusion

In this article, we proposed and developed an IoT-enabled scalable closed hydroponics system that can monitor pH, EC, water temperature, air temperature, humidity, and CO₂ concentration and control them automatically. We also proposed an algorithm to reduce instances of passing erroneous sensor data to classification methods to prevent systems from undertaking unnecessary or even detrimental control actions. To our knowledge, no such algorithm has been comprehensively implemented in an IoT-enabled hydroponic system, as we proposed. Mitigating erroneous sensor data also heavily reduces the need for human intervention in such systems, which can get exhaustive at times since systems that must be run around the clock also call for constant supervision and vigilance. The intelligent classification method also builds on an XGBoostClassifier model and can accurately determine the appropriate control action needed based on the sensor data.

To meet the growing global caloric demand, newer methods of agriculture need to be implemented to either replace or complement current conventional

745 techniques, which are soil-based. With the diminishing availability of arable
746 land and deterioration in the quality of already available farmland, the need
747 to implement hydroponic farming at a larger scale grows ever more perti-
748 nent. To that extent, our proposed system, coupled with the error resilience
749 algorithm and intelligent control action classification method, may show a
750 way toward a sustainable global food supply, ensuring food security across
751 all levels, from international to local.

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758 9. Appendices

759 The code files and data acquired from the sensors can be found in the
760 link below:

761 [https://github.com/MSIChowdhury/Design-of-an-IoT-enabled-Scalable-Closed-](https://github.com/MSIChowdhury/Design-of-an-IoT-enabled-Scalable-Closed-Hydroponics-System-with-Smart-Parameter-Control)
762 [Hydroponics-System-with-Smart-Parameter-Control](https://github.com/MSIChowdhury/Design-of-an-IoT-enabled-Scalable-Closed-Hydroponics-System-with-Smart-Parameter-Control)

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